Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 17: Sorting Into Boxes

Equivalence Classes and the Partition Theorem

Chartrand & Zhang, Section 5.2

Name: _____ Date: _____

Two ideas about to meet

In Lecture 14 you met **partitions**: a way of cutting a set into blocks so that every element lands in exactly one. On Wednesday you met **equivalence relations**.
Today's theorem says these are *the same idea*. Every equivalence relation produces a partition, and every partition comes from an equivalence relation.

By the end of today you should be able to

1. Compute the equivalence classes of a relation.
2. State why two classes are always equal or disjoint.
3. Explain the correspondence between equivalence relations and partitions.

1. Equivalence classes

Definition 5.17

Let R be an equivalence relation on a set A. For $a \in A$, the **equivalence class** of a is

$$[a] = \{x \in A : x \, R \, a\}.$$

Think of $[a]$ as the set of all the *relatives* of a.

Since R is reflexive, $a \, R \, a$, so $a \in [a]$. Therefore no equivalence class is empty.

Example 5.18. On $S = \{1, 2, 3, 4, 5, 6\}$, take

$$R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6), (1,4), (2,3), (2,6), (3,2), (3,6), (4,1), (6,2), (6,3)\}.$$

$[1] = \{1, 4\}$	$[2] = \{2, 3, 6\}$	$[3] = \{2, 3, 6\}$
$[4] = \{1, 4\}$	$[5] = \{5\}$	$[6] = \{2, 3, 6\}$

Six elements, but only three *distinct* classes: $\{1, 4\}$, $\{2, 3, 6\}$, $\{5\}$.

Notice $[1] = [4]$ and $[2] = [3] = [6]$. Different labels, same box.

2. Classes on the integers

For $a \; R \; b$ if $a + b$ is even (Wednesday's relation):

$$[0] = \{x \in \mathbb{Z} : x + 0 \text{ is even}\} = \text{the even integers}$$
$$[1] = \{x \in \mathbb{Z} : x + 1 \text{ is even}\} = \text{the odd integers}$$

And $[4] = [0]$, $[-3] = [1]$. Two distinct classes in all.

You have seen this partition before

Evens and odds. That was the first example of a partition in Lecture 14, and it is the partition of $\mathbb{Z}$ produced by this relation.
Try the relation $a \; R \; b$ if $a \equiv b \pmod 3$ and you get three classes — the integers leaving remainder 0, 1, or 2.

3. The key theorem

Theorem 5.22 and Corollary 5.24

Let R be an equivalence relation on A, and let $a, b \in A$.

$$\text{If } a \; R \; b \text{ then } [a] = [b]. \qquad \text{Otherwise } [a] \cap [b] = \emptyset.$$

So any two equivalence classes are **either equal or disjoint**. Never partly overlapping.

Theorem 5.23

The distinct equivalence classes of an equivalence relation on A form a **partition** of A.

Why it works — match each partition requirement to the property that supplies it.

Partition requires	Supplied by
Every block is nonempty	Reflexivity: $a \in [a]$.
Every element is in some block	Reflexivity again: $a \in [a]$.
Blocks do not partly overlap	Symmetry and transitivity together.

The overlap argument, in one line

Suppose $x \in [a]$ and $x \in [b]$. Then $x \; R \; a$ and $x \; R \; b$. By symmetry, $a \; R \; x$; combining with $x \; R \; b$ and using transitivity gives $a \; R \; b$, so $[a] = [b]$.
A shared element forces the whole boxes to coincide. **That is why partial overlap cannot happen.**

4. The correspondence runs both ways

Both directions

Relation → partition. The distinct equivalence classes are the blocks. (Theorem 5.23.)
Partition → relation. Given a partition, define $a \ R \ b$ to mean *a and b lie in the same block.* That relation is reflexive, symmetric, and transitive.

Check the second direction. With R meaning "same block":

Reflexive? Yes — a is in the same block as itself.
Symmetric? Yes — same block is a symmetric condition.
Transitive? Yes — if a, b share a block and b, c share a block, it is the same block.

Why this matters

An equivalence relation and a partition are two descriptions of one situation. Whenever you sort objects into groups where every object lands in *exactly one* group — files by extension, integers by remainder, students by section — you have both.
That is why these three properties, and not some other three, are the ones with a name.

5. Going backwards

Let $S = \{a, b, c, d, e\}$ and suppose the distinct equivalence classes are $\{a, c\}$ and $\{b, d, e\}$. Write out R.

Your work

$R = \{(a,a), (a,c), (c,a), (c,c),\ (b,b), (b,d), (b,e), (d,b), (d,d), (d,e), (e,b), (e,d), (e,e)\}$ — every pair within a block, and none across blocks.

How many pairs? A block of size k contributes k^2 pairs, so here $2^2 + 3^2 =$ 13.

Exit ticket

1. For $a \ R \ b$ if $|a| = |b|$ on $\mathbb{Z}$, describe $[3]$ and $[0]$.

2. Can two distinct equivalence classes share exactly one element? Explain.

Before Monday

Read Section 5.3, "Functions," pp. 158–164. A function is a special kind of relation, so this builds directly on today.

Self-check (not collected). Section 5.2, Exercises 5, 9, and 13 — odd-numbered, answers in the back.

Coming up

Homework 6 is due tonight, Friday, September 25, 11:59 p.m.
Quiz 6 is Monday, September 28 — 15 minutes on Sections 5.1 and 5.2.
Homework 7 covers Sections 5.1 and 5.2 and is due Friday, October 2.

Questions I still have